

Create Virtual CPUs Using CPU Sim

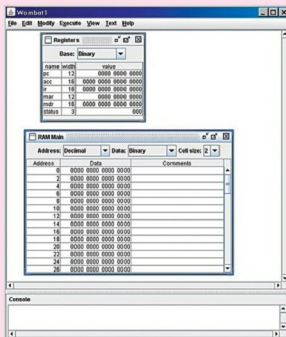
CPU Sim is a software that lets you create CPU architectures of your choice, create and edit assembly languages, and run them on the simulated CPU using machine language

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Understanding the concepts of computer organisation and architecture is difficult by just reading about it. Hands-on experience is required to fully grasp the subject. Unfortunately, providing this experience can be difficult due to the lack of availability and high cost in setting up labs with necessary hardware. Moreover, the time required by the users to familiarise with the tools for working with hardware is also significant. In order to solve this problem, some central processing unit (CPU) simulator programs were developed. But most of these software packages could simulate only fixed architectures.

Why you should learn to use CPU Sim

The idea that you should have exposure to as many different architectures as possible and hands-on experience to design your very own simple architectures is what led to the development of CPU Sim. It is an interactive simulator that allows you to understand computer architectures and organisations, design your own architectures from scratch, or modify the given architectures. You can write the programs using machine language or Assembly language and run them on the virtual CPU. You can also specify details of the CPU to be simulated, such as register set, microinstruction set, machine instruction set, memory and Assembly language instructions. The software was based



Main window

on STARTLE simulator developed by JM Kerridge of the University of Sheffield.

What the user gets

By using CPU Sim, you can get hands-on experience on the following subjects of computer organisation and architecture:

1. Representation of numeric data and their number bases
2. Non-numeric data representation
3. Microprogrammed realisation of the CPU (where control signals are generated by a microcode/microprogram stored in the control read-only memory)
4. Operation of control unit: in-

struction fetch, decode and execution

5. Instruction types (data manipulation, memory architecture, registers, interrupt, control, external I/O)

6. Instruction formats

7. Assembly language programming and machine language programming

8. Addressing modes of CPU

9. Main memory organisation and operations

Understanding and using CPU Sim does not require in-depth knowledge of a high-level language, but an appreciation of concepts involved in high-level language

programming is an advantage.

What you can do with it

CPU Sim is a complete development environment. This software allows creating or modifying a simulated CPU including units such as registers, microinstructions, machine instructions and RAM, and specifying its features.

You can create, edit and execute Assembly language programs for this virtually created CPU. It has built-in features and dialogue boxes that help you design CPU architecture, text editor where you can write and edit the Assembly language program, assembler and separate displays for